

Introducció als tipus d'estudis ("quantitative studies")

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Qualitative vs Quantitative Methods

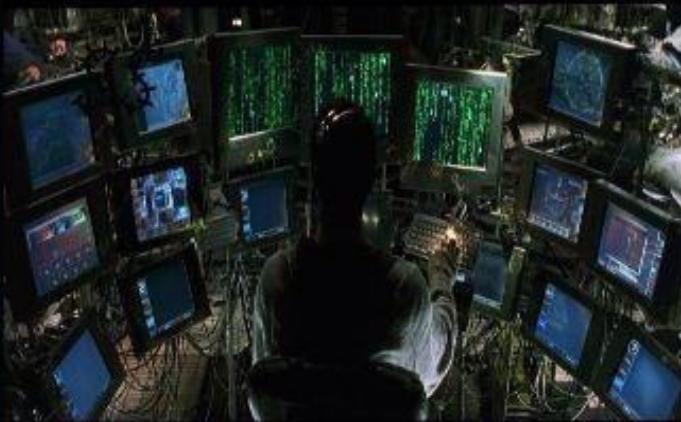
Qualitative Methods

1. Methods include focus groups, in-depth interviews, and **reviews** of documents for types of themes
2. Primarily **inductive process** used to formulate theory or hypotheses
3. **More subjective**: describes a problem or condition from the point of view of those experiencing it
4. **Text-based**
5. More in-depth information on a **few cases**
6. **Unstructured or semi-structured response** options
7. **No statistical tests**
8. Can be valid and reliable: largely depends on skill and rigor of the researcher
9. **Less generalizable**

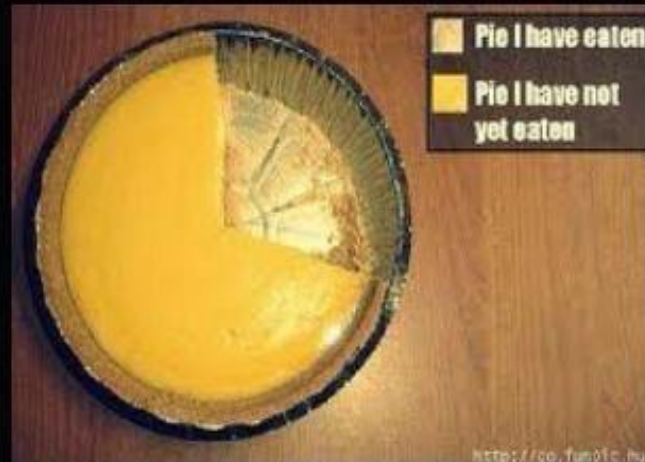
Quantitative Methods

1. Structured observations, and reviews of records or documents for **numeric information**
2. Primarily **deductive process** used to test pre-specified concepts, constructs, and hypotheses that make up a theory
3. **More objective**: provides observed effects (interpreted by researchers) of a program on a problem or condition
4. **Number-based**
5. Less in-depth but more breadth of information across **a large number of cases**
6. **Fixed response** options
7. **Statistical tests** are used for analysis
8. Can be valid and reliable: largely depends on the measurement device or instrument used
9. **More generalizable**

STATISTICIAN



What my friends think I do



What my mom thinks I do



What society thinks I do



What my boss thinks I do

	wtbyht	ver
18	23.09	
19	22.15	
20	20.36	
21	20.00	
22	22.73	
23	23.64	480.00 600.00 2.00

What I think I do



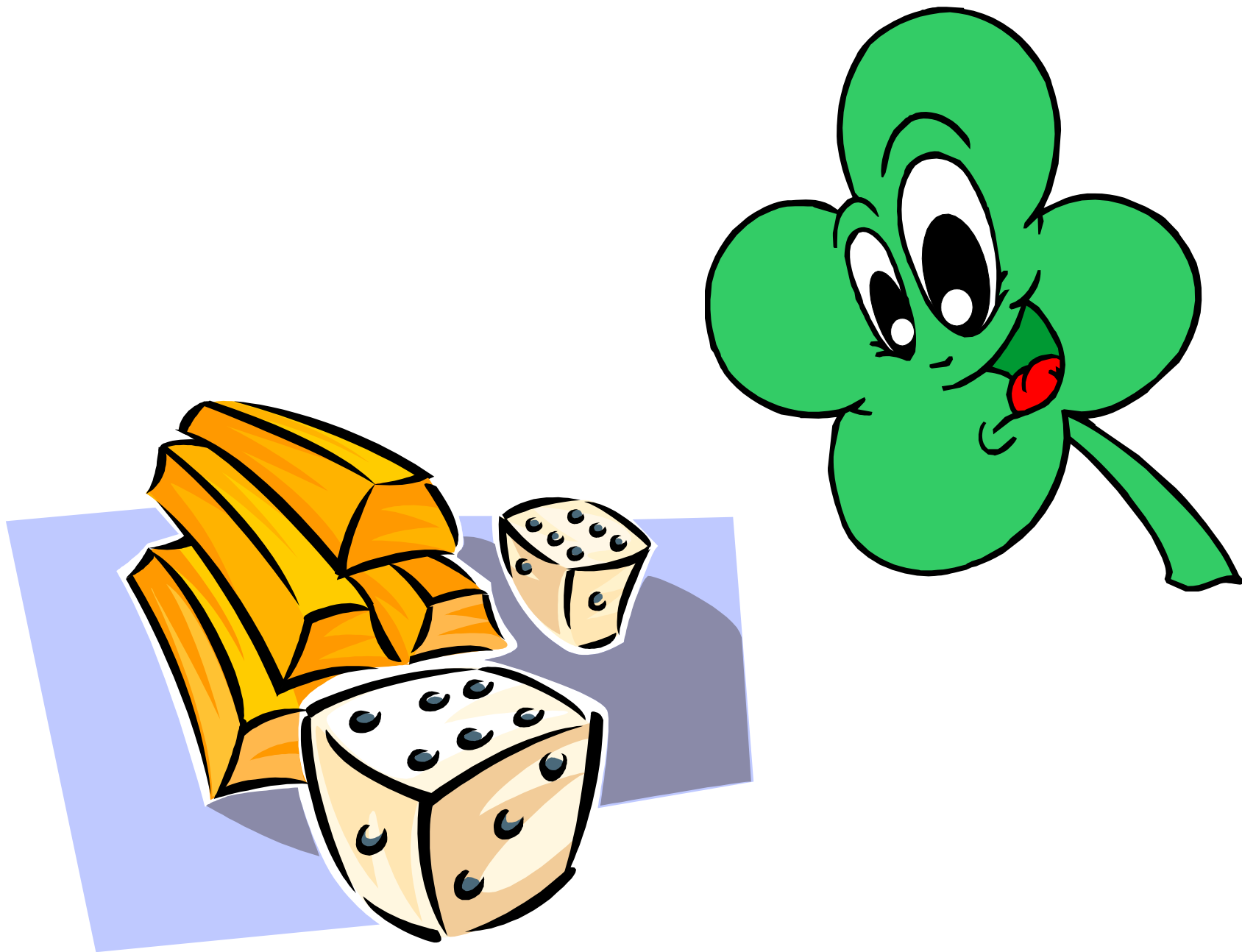
What I actually do



P value, is the probability...

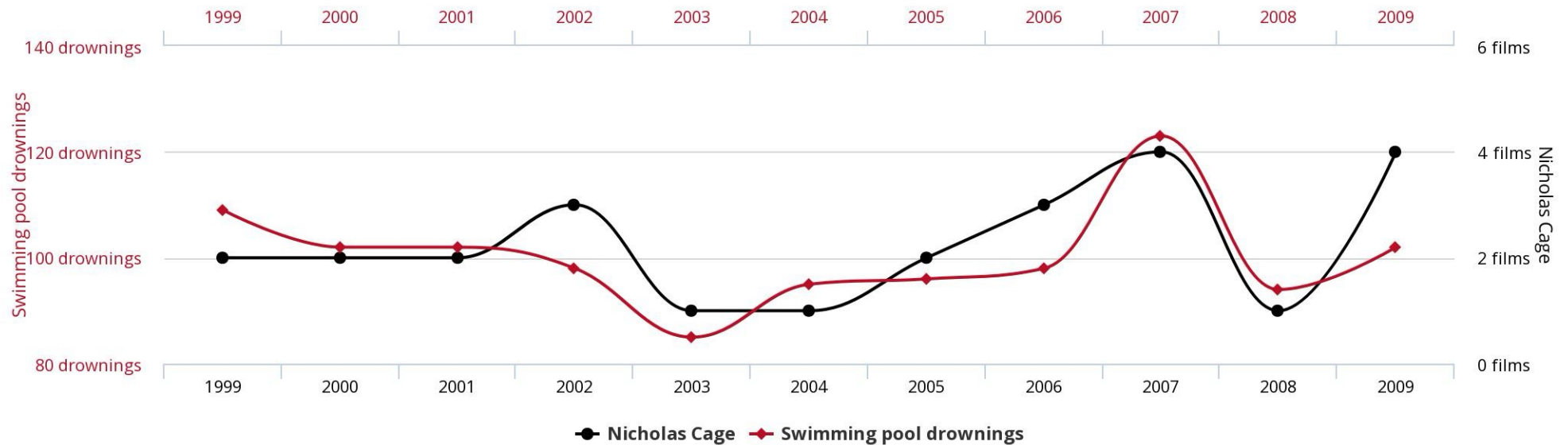
- a) that the null hypothesis is true
- b) that the alternative hypothesis is true
- c) that the null hypothesis is false
- d) that the alternative hypothesis is false
- e) of the given result or more extreme results if the null hypothesis were true

<https://doi.org/10.1136/bmj.a201>



Spurious correlations

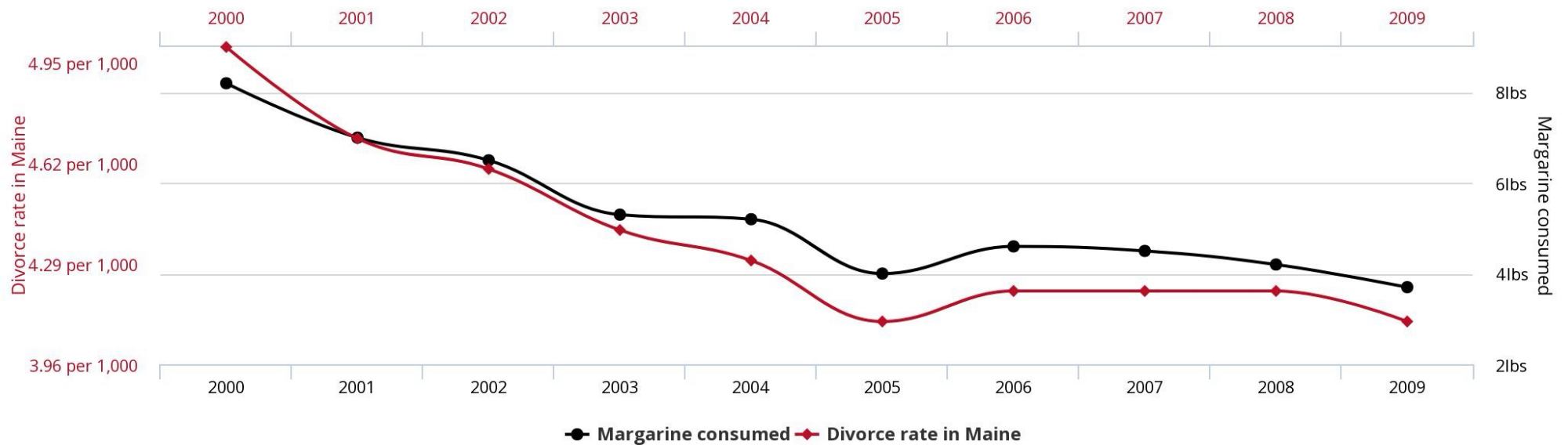
Number of people who drowned by falling into a pool
correlates with
Films Nicolas Cage appeared in



tylervigen.com

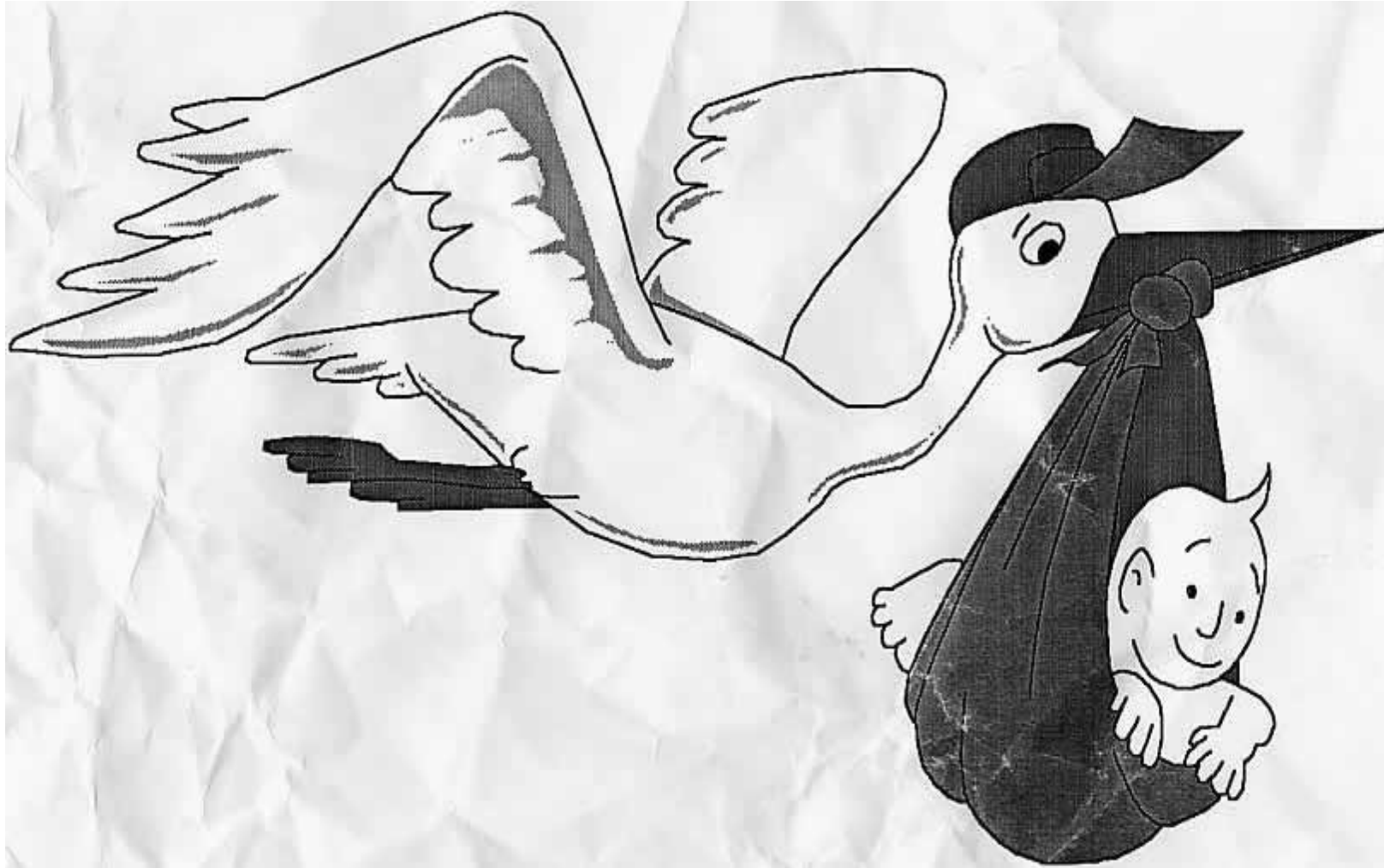
Spurious correlations

Divorce rate in Maine correlates with Per capita consumption of margarine

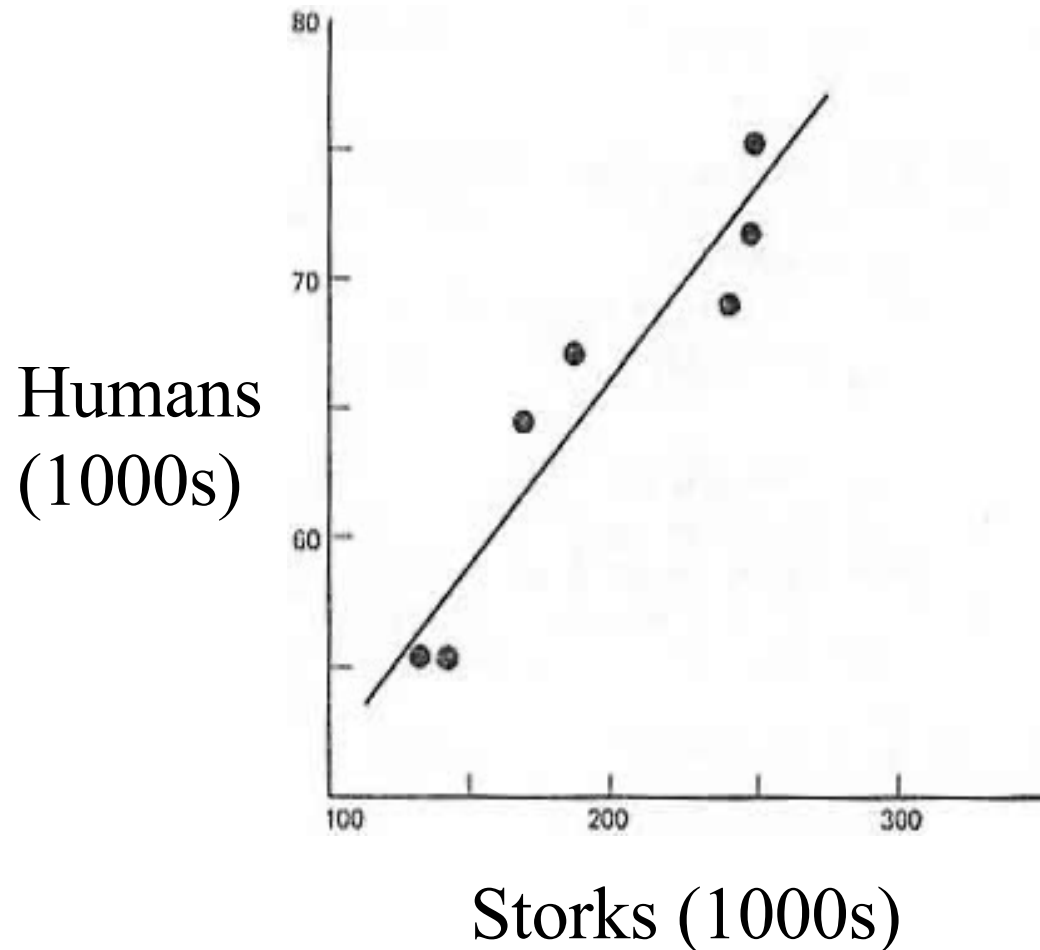


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Do Storks Bring Babies?



Population of Oldenburg, Germany, 1930-1936
(*Ornithologische Monatsberichte* 44, Jahrgang, 1936, Berlin)



- ***“Association does not mean causation”***

- Sir Austin Bradford Hill's criteria:

1. Strength of association
2. Consistency
3. Specificity
4. Temporal sequence
5. Biological gradient
6. Biological plausibility
7. Coherence
8. Experiment
9. Analogy

Hill AB

The Environment and Disease: Association or Causation?
Proc R Soc Med. 1965;58:295-300

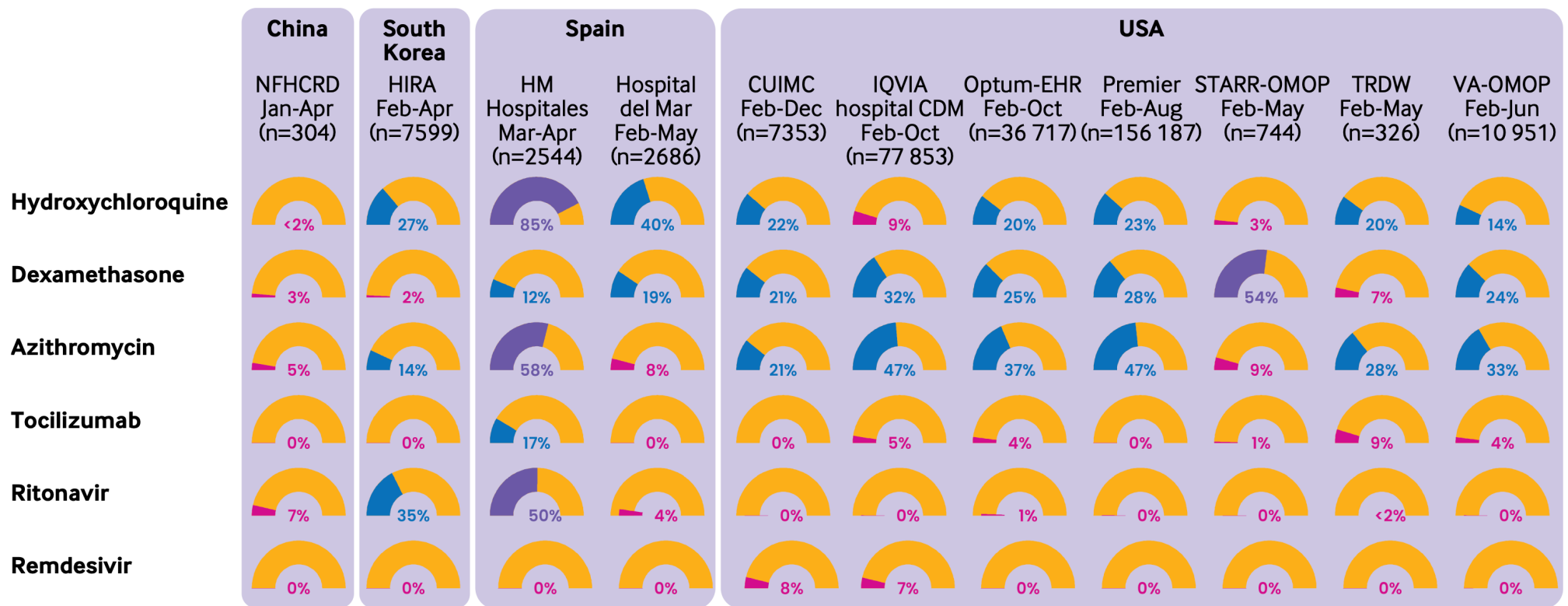


Fig 3 | Percentage of any use (one day or more) of selected drugs on days 0 to 30 after hospital admission in patients with a positive test result for or diagnosis of covid-19. CUIMC=Columbia University Irving Medical Center; HIRA=Health Insurance Review and Assessment; OMOP=Observational Medical Outcomes Partnership; Optum-EHR=Optum deidentified electronic health record dataset; STARR=Stanford medicine Research data Repository; TRDW=Tufts Research Data Warehouse; VA=Veterans Affairs

Prats-Urbe A. BMJ 2021 May 11;373:n1038

SALUD

Hidroxicloroquina: el fármaco que nunca debió usarse masivamente contra el Covid

Un estudio estima que el medicamento podría haber causado casi 17.000 muertes en seis países, 2.000 de ellas en España. Los expertos señalan que lo sucedido con esta terapia durante la pandemia es un ejemplo de lo que no debería volver a ocurrir

<https://www.elmundo.es/ciencia-y-salud/salud/2024/01/17/65a537b8fdddf54b58b45b4.html>

Statistics Notes

Absence of evidence is not evidence of absence

Douglas G Altman, J Martin Bland

BMJ 1995;311:485



Table I. Probability of statistical significance ($p < 0.05$) upon repetition of an experiment as a function of the p -value of the first experiment, under two different distributions for the true effect size μ (see text)

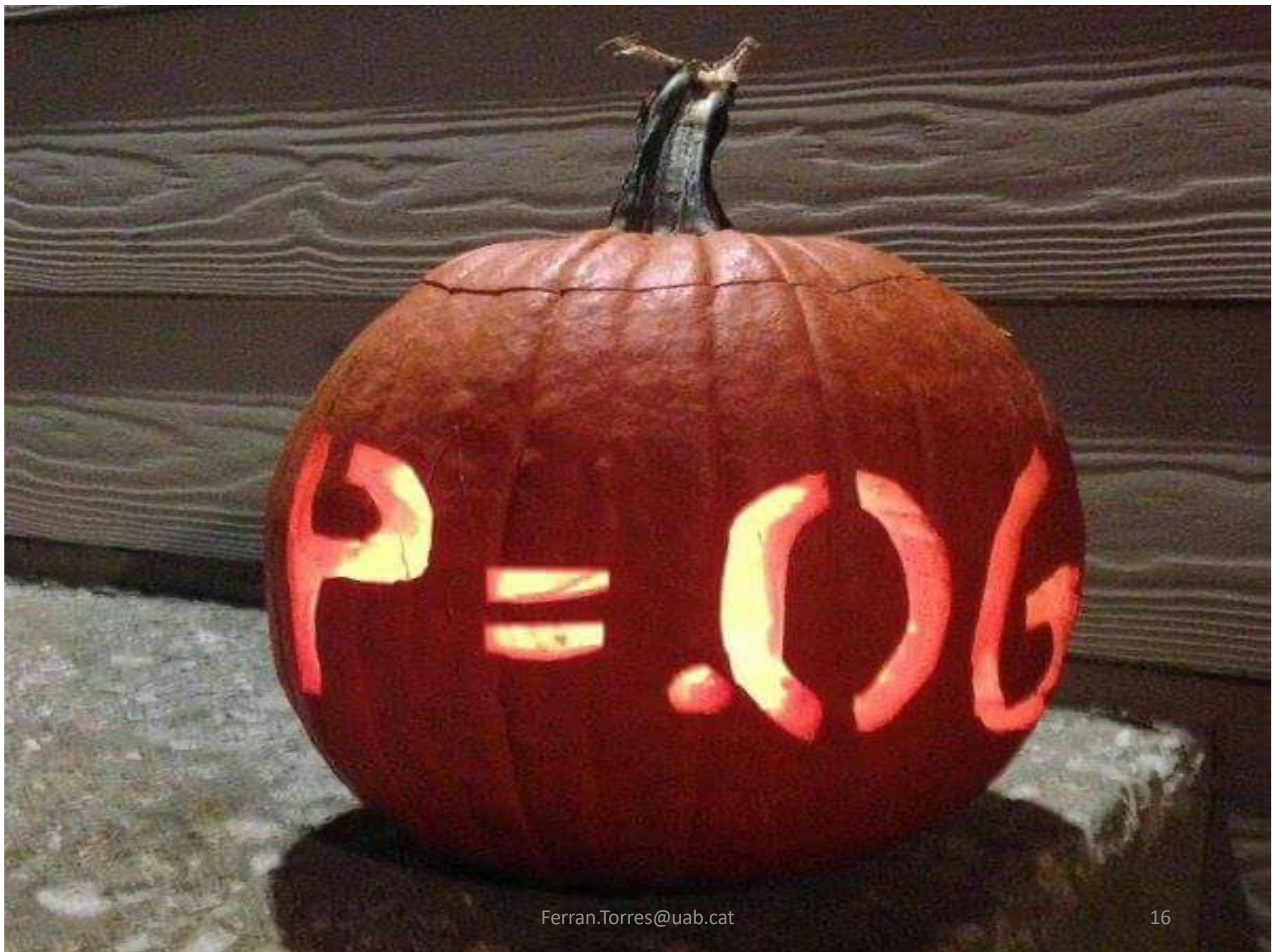
Probability of a statistically significant result in duplicate experiment when:

P -value of initial experiment	μ equals the difference x observed in the first experiment	μ has a uniform prior distribution before the first experiment
0.10	0.37	0.41
0.05	0.50	0.50
0.03	0.58	0.56
0.01	0.73	0.66
0.005	0.80	0.71
0.001	0.91	0.78

Goodman SN.

A comment on replication, p -values and evidence.

Stat Med 1992;11:875-9



Randomized Trials, Statistics, and Clinical Inference

Gregg W. Stone, MD,* Stuart J. Pocock, PhD†

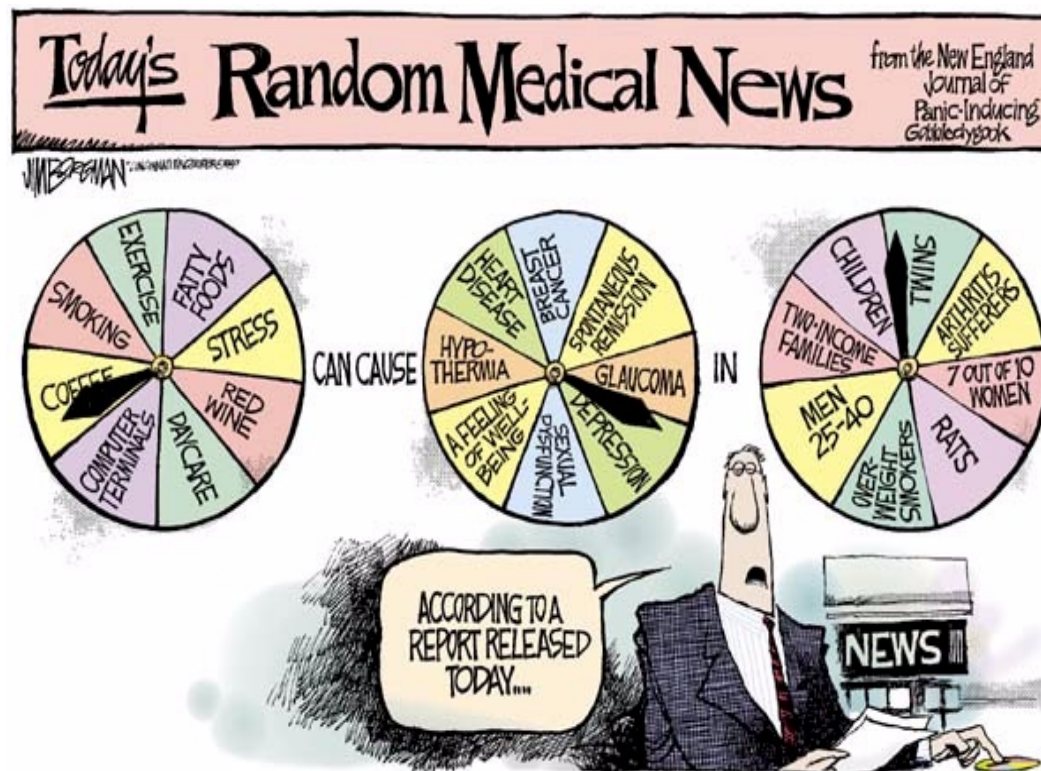
New York, New York; and London, United Kingdom

“A p value is no substitute for a brain.”
—Anonymous

Stone GW & Pocock SJ.
Randomized trials, statistics, and clinical inference.
J Am Coll Cardiol. 2010;55:428-31

Three premises

1. it is important to adhere to a prespecified analysis plan (if one exists)



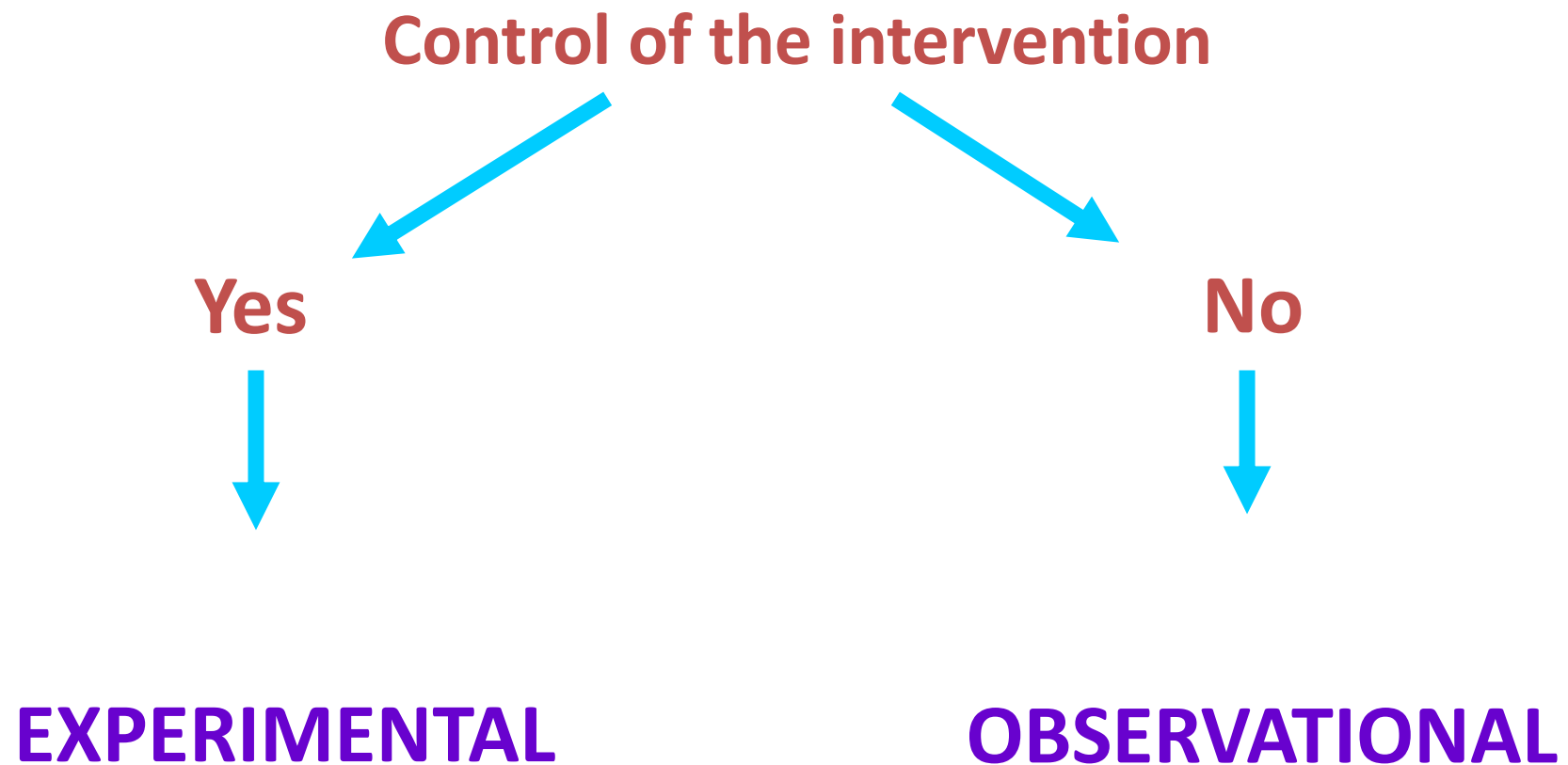
Harrington D et al.
New Guidelines for Statistical Reporting in the *Journal*.
NEJM 2019;381:285-286

Three premises

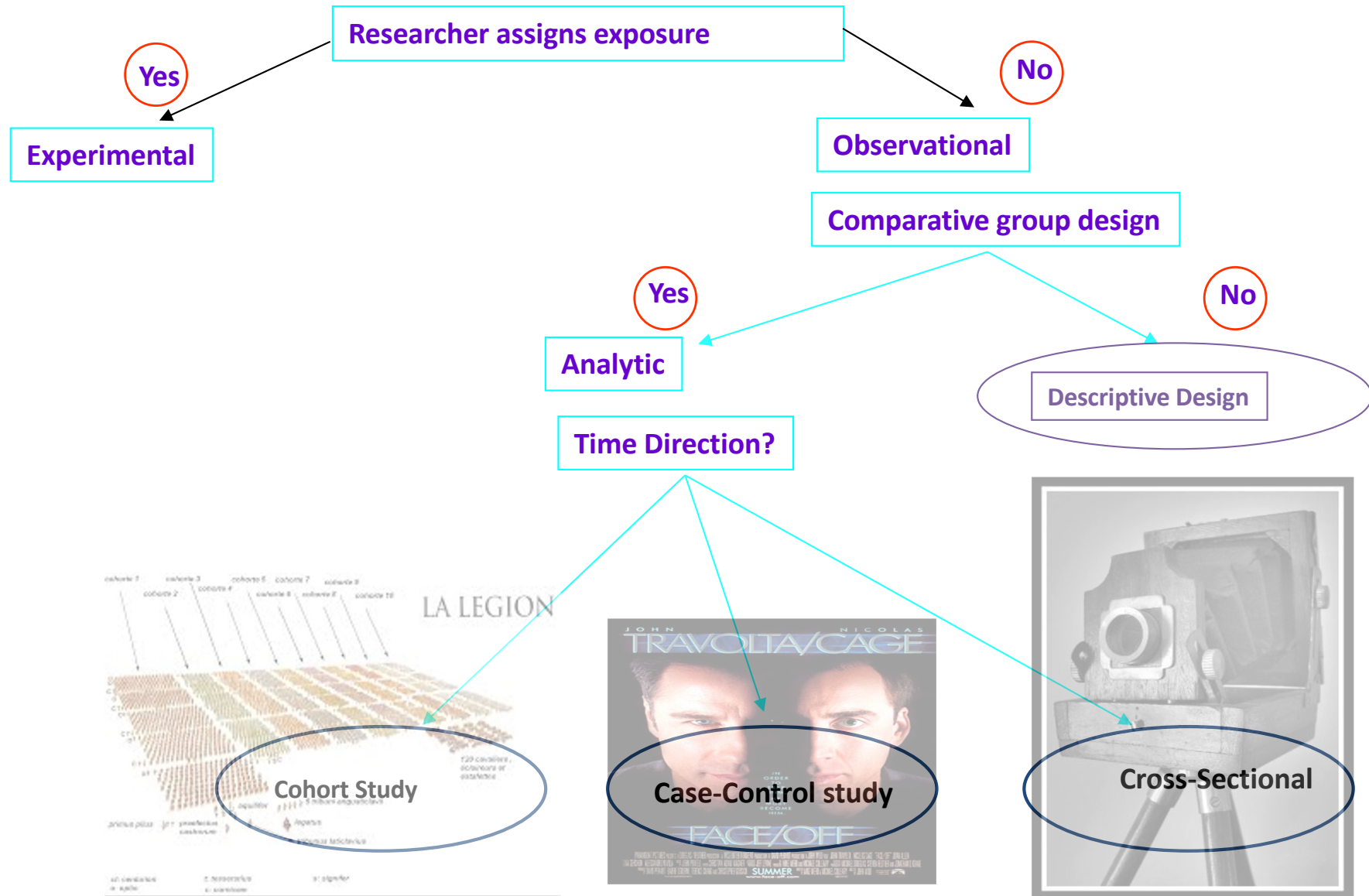
1. it is important to adhere to a prespecified analysis plan (if one exists)
2. the use of statistical thresholds for claiming an effect or association should be limited to analyses for which the analysis plan outlined a method for controlling type I error
3. and the evidence about the benefits and harms of a treatment or exposure should include both point estimates and their margins of error

Harrington D et al.
New Guidelines for Statistical Reporting in the *Journal*.
NEJM 2019;381:285-286

Selection of the quantitative design

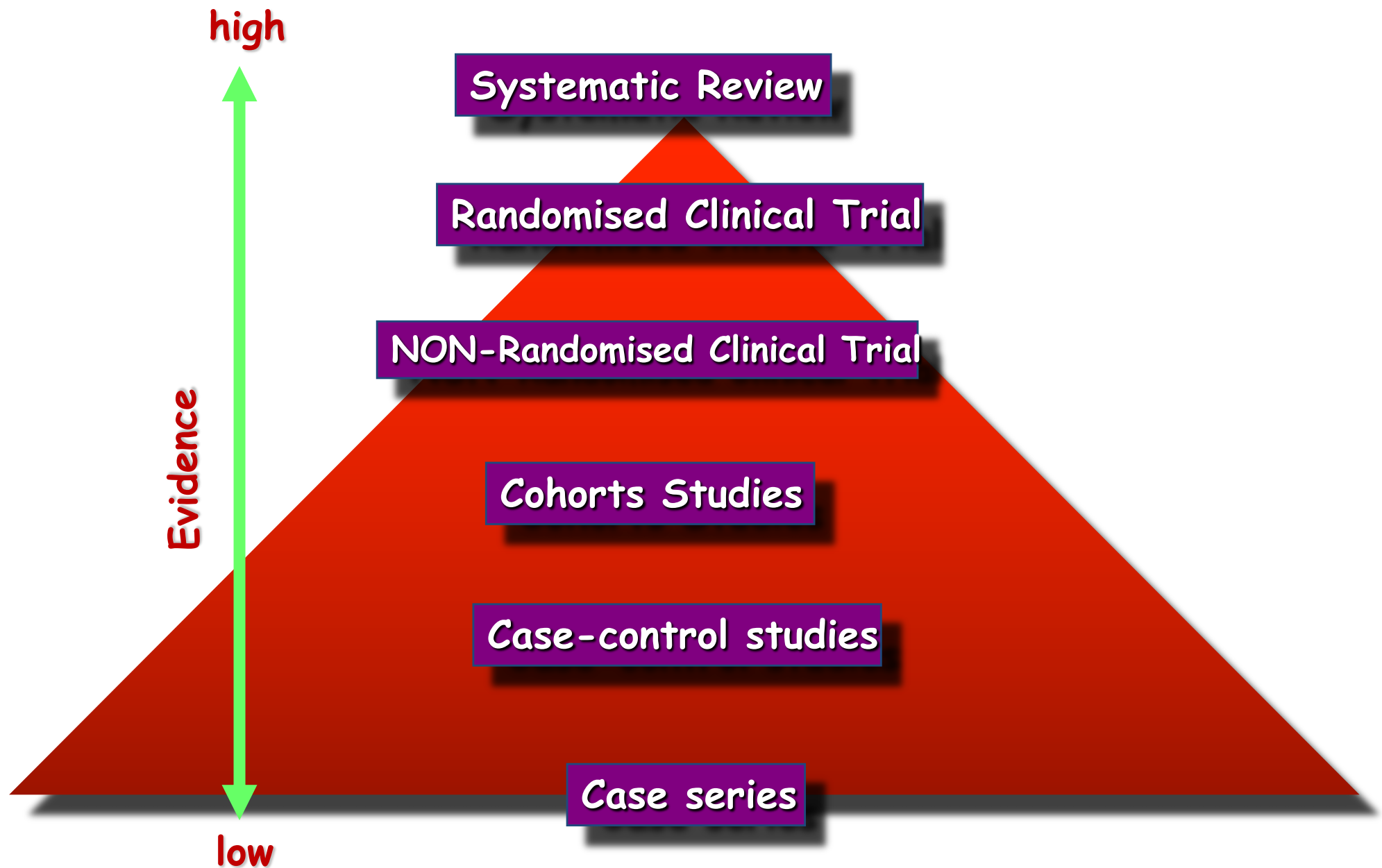


Main study design

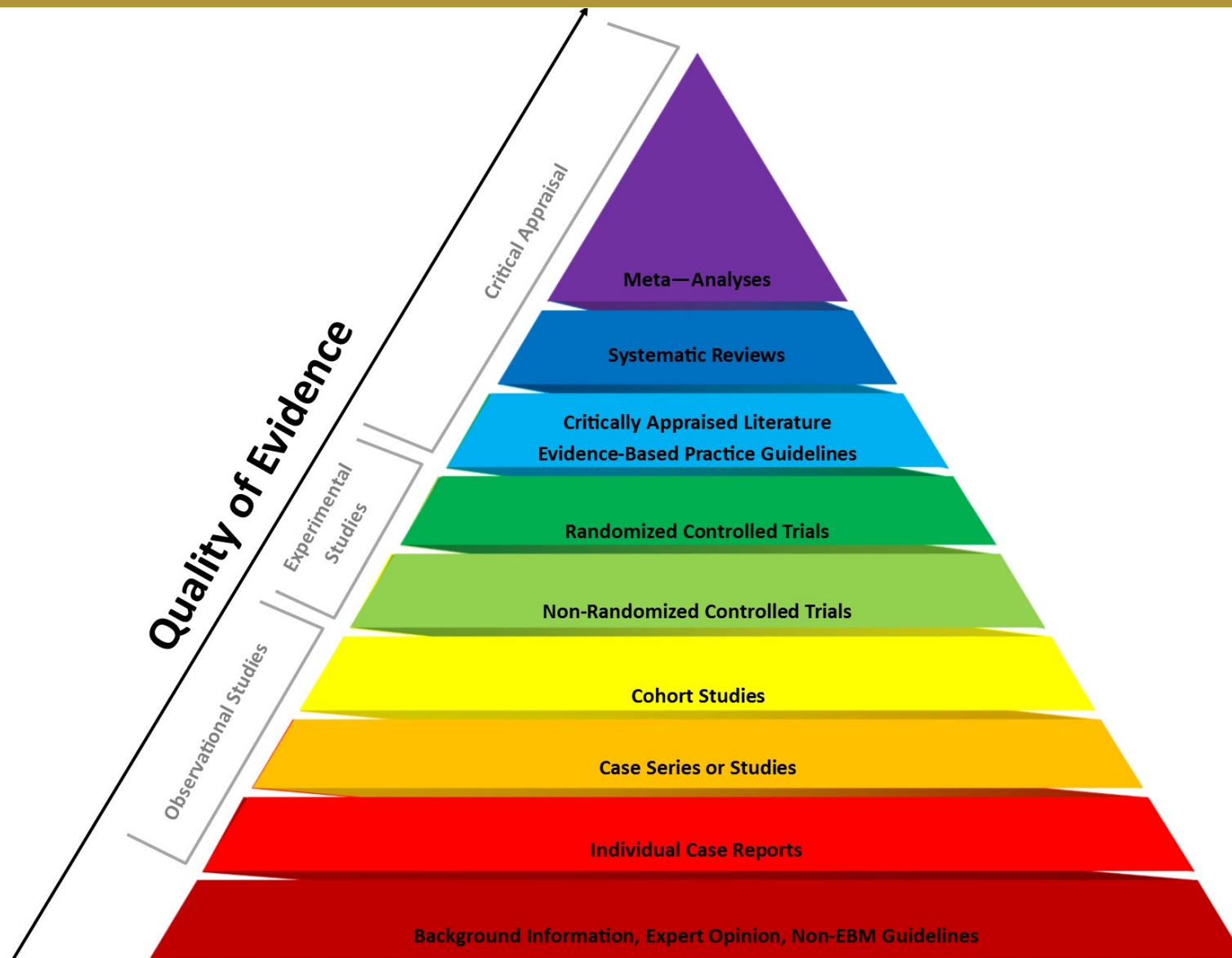


Ensayo Clínico Aleatorio



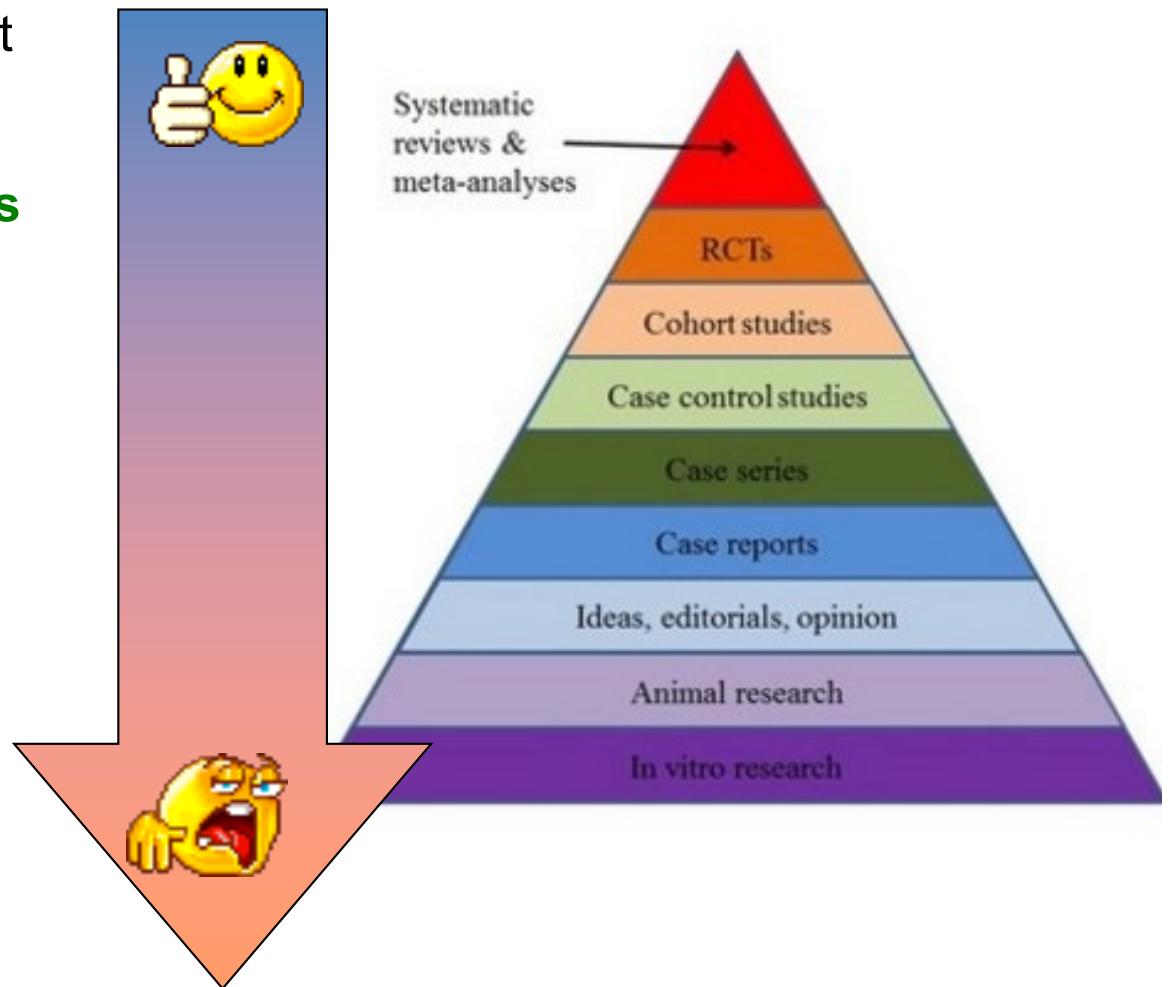


On the evidence for decisions about the use of therapeutic interventions



“Hierarchies” of evidence

- Meta-analysis of randomized controlled trials with consistent results
- **Randomized controlled trials**
- observational studies
- Collections of cases and anecdotal cases
- Expert Opinions
- Animal research
- Research in vitro



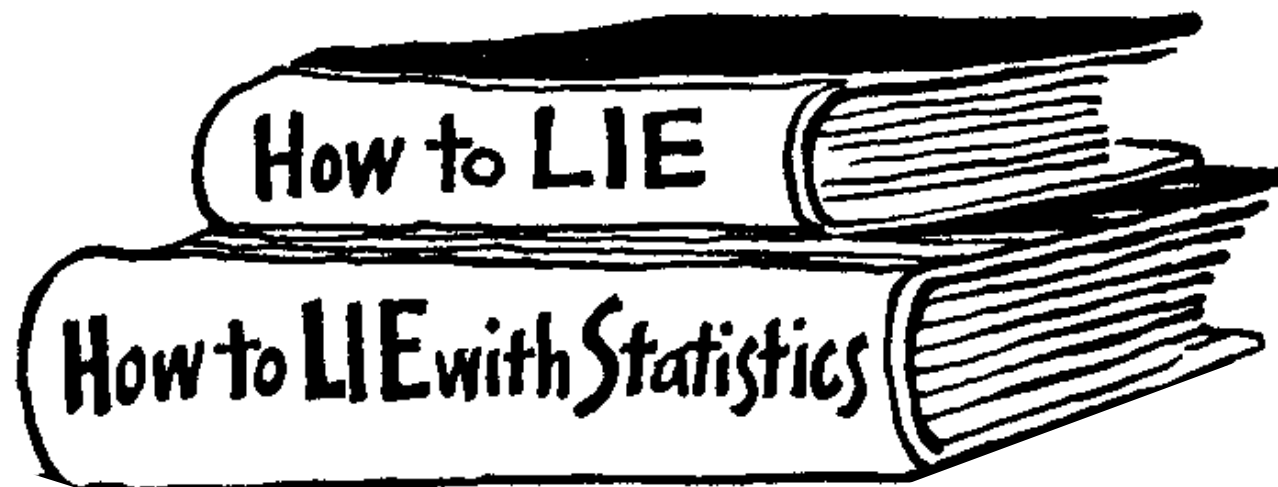
The role of statistics

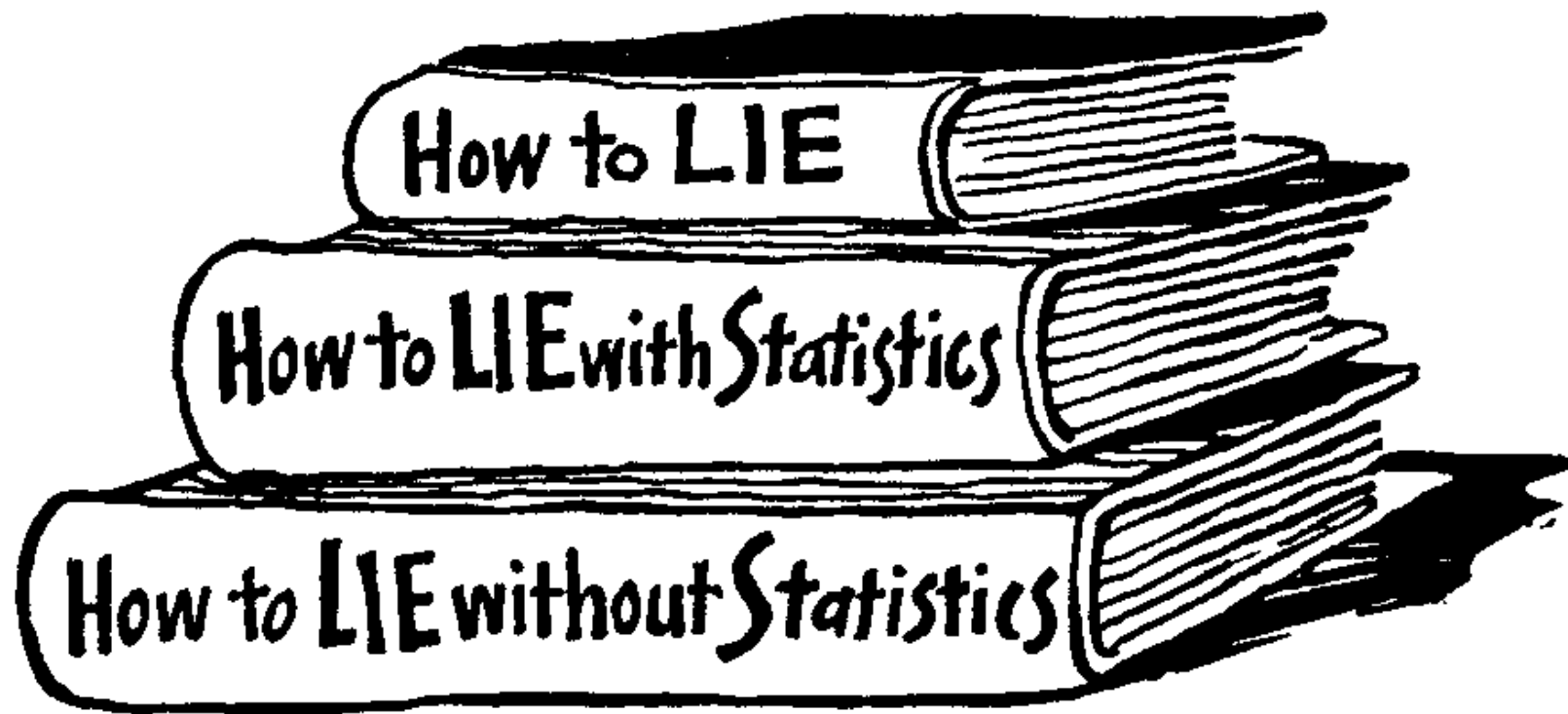


“Thus statistical methods are no substitute for common sense and objectivity. They should never aim to confuse the reader, but instead should be a major contributor to the clarity of a scientific argument.”

The role of statistics. Pocock SJ . *Br J Psychiat* 1980; 137:188-190







A close-up photograph of a ginger cat with large, dark, expressive eyes. The cat is looking directly at the camera with a calm expression. Its fur is a warm orange-brown color, and its whiskers are long and white. The background is a soft, out-of-focus green, suggesting foliage. The text "Thanks!!!" is overlaid in the top right corner in a white, bold, sans-serif font.

Thanks!!!